CASE OVERVIEW

LS / Supply Chain
Predictive maintenance Approach

Deviation Management & CAPA in GMP Manufacturing

AI-powered agents assist in real-time deviation handling by analyzing events, retrieving relevant SOPs and historical cases, and generating CAPA (Corrective and Preventive Actions) recommendations, reports, and audit trails.

Business Challenges

- Manual deviation analysis is slow and inconsistent (30–60 min per event)
- Operators must manually search SOPs and historical records
- CAPA decisions depend on individual experience
- Audit trail creation is manual and error-prone
- High risk of missed steps under time pressure
- Inspection readiness requires significant manual effort

Use Case Description

PROBLEM STATEMENT

In life sciences (pharma/biotech/medical devices), GMP is the set of regulated quality standards that ensure products are consistently made and controlled so they are safe, effective, and meet specifications. Operations teams must keep assets and batches within validated limits while juggling equipment health, preventive maintenance, and strict SOP/BPR compliance, so monitoring assets in real time, detecting early failure signals, optimizing "golden batch" parameters, and identifying potential deviations fast without disrupting production

OVERVIEW

Using Microsoft AI Foundry, an Operations Agent orchestrates agents that stream real-time sensor/SCADA data to detect early wear patterns and cross-check them against SOP/BPR "golden batch" limits.
Example: a vibration spike on a granulator triggers a predicted bearing failure alert plus a recommended PM plan and deviation-resolution playbook retrieved from validated SOPs. AI Foundry supports it with agent orchestration, RAG over controlled documents, evaluation, and governed deployment.

WHY

KPIs Impacted

- Decision Time ↓ (30+ min → <5 min)
- Manual QA Effort ↓ (auto-drafted reports)
- Deviation Handling Errors ↓
- Compliance Consistency ↑
- Inspection Readiness Time ↓

Business Impact

- Faster deviation triage and resolution
- Reduced batch delays and rework costs
- Standardized GMP-compliant decision-making
- Improved audit readiness and traceability
- Reduced reliance on operator expertise

HOW

Functional Solution

- Detect anomaly events from MES / SCADA / IoT
- Automatically build structured incident context
- Context-aware analysis
- AI agents analyze deviation and validate against SOP
- Generate CAPA recommendations based on historical cases
- Produce structured report and audit trail
- Human reviews and approves actions

Technical Solution

- Event ingestion via Azure Functions
- Context enrichment from MES, CMMS, BPR systems
- RAG-based retrieval via Azure AI Search (SOP, CAPA history)
- Agents implemented in Azure AI Foundry Agent Service
- Integration with enterprise systems (QMS / CMMS)
- Full audit logging for compliance

WHO

Stakeholders

- Production operators
- QA / Quality engineers
- Compliance & audit teams
- QA managers
- IT / Digital transformation teams

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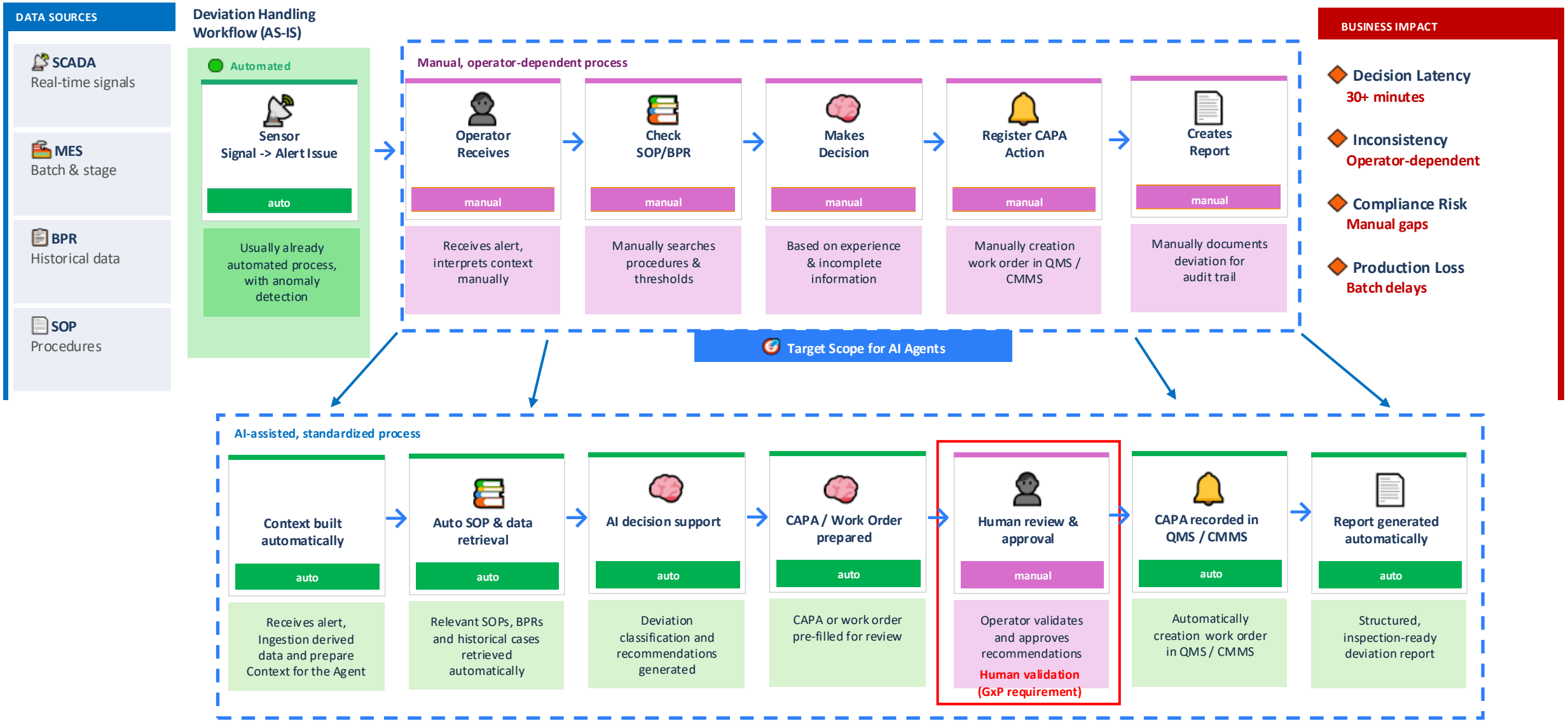
Use Case Scope in the Business Process



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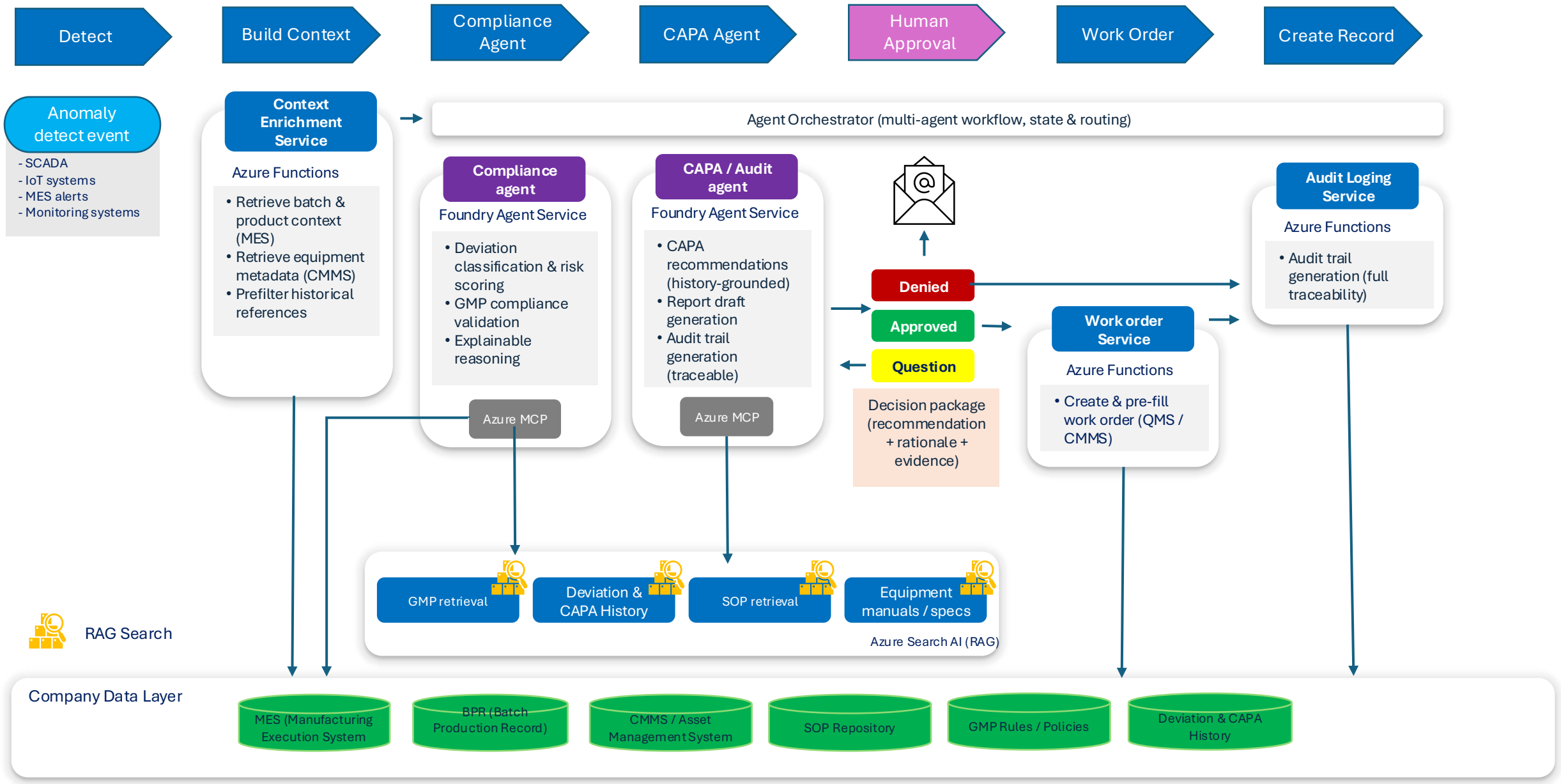
System Architecture



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